Topic Modeling and Contextual Analysis of Mental Health Discourse in Indian Parliamentary Debates

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This dissertation applies BERTopic neural topic modeling to a curated corpus of Rajya Sabha parliamentary debates on mental health spanning 2000–2025, constructed via a seven-phase multilingual NLP pipeline.

Abstract

Mental health has emerged as a significant axis of public policy concern in India, yet the legislative debates surrounding it remain analytically underexplored at scale. This dissertation applies **BERTopic**—a transformer-based neural topic modeling framework—to a large, purpose-built corpus of Indian Parliamentary debates sourced from the Rajya Sabha archive spanning 2000 to 2025.

The study constructs an end-to-end data engineering pipeline that harvests archival parliamentary records from the official DSpace repository, processes both machine-readable and scanned (OCR) documents, applies heuristic text normalization with language-aware Hindi protection, performs Unicode-block-based multilingual language detection, machine-translates all non-English content via the Google Translate API, and filters the resulting text to a domain-specific, English-only corpus primed for downstream Natural Language Processing (NLP). This seven-phase pipeline yields a semantically dense collection of parliamentary utterances directly pertaining to mental healthcare policy.

BERTopic is applied to this corpus to identify latent thematic clusters, trace their temporal evolution across parliamentary sessions, and examine how legislative attention to mental health topics has shifted in relation to key policy milestones—the National Mental Health Policy (2014) and the Mental Healthcare Act (2017). The performance of BERTopic is additionally evaluated against classical Latent Dirichlet Allocation (LDA) on standard topic coherence metrics.

The findings contribute to computational social science and mental health policy research by demonstrating how neural NLP can illuminate legislative priorities, quantify thematic salience, and reveal shifts in governmental discourse on mental health governance in India over a 25-year period.

Keywords: BERTopic, Latent Dirichlet Allocation, Topic Modeling, Mental Health Policy, Indian Parliament, Rajya Sabha, Natural Language Processing, Multilingual NLP, Parliamentary Discourse, Computational Social Science.

Declaration

I hereby declare that this dissertation, submitted in partial fulfilment of the requirements for the degree of Bachelor of Science (Honours) in Quantitative Economics and Data Science at the Birla Institute of Technology, Mesra, is my own original work. All sources consulted have been duly acknowledged. This work has not been submitted, in whole or in part, for any other degree or professional qualification.

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Chapter 1

Introduction

1.1 Background and Motivation

Parliamentary debates occupy a privileged position in democratic governance. They constitute the formal, public record of a nation's legislative priorities, contested values, and policy trajectories—the arena in which elected representatives negotiate the terms of collective life. Within the Indian Parliament, and in particular the Rajya Sabha (Upper House), Members of Parliament deliberate on a vast spectrum of issues from economic legislation to social welfare, generating an enormous repository of unstructured, highly multilingual textual data spanning decades.

Mental health has, in recent years, emerged as a significant public policy concern in India. The World Health Organization estimates that nearly 7.5 per cent of India's population lives with some form of mental disorder, representing one of the highest disease burdens globally (WHO, 2022). Despite this epidemiological weight, mental health legislation in India has historically been fragmented and under-resourced. For over a century, the legislative landscape was governed by the Indian Lunacy Act of 1912—a colonial-era statute rooted in custodial and punitive conceptions of mental illness. Landmark reforms, including the National Mental Health Policy of 2014 and the Mental Healthcare Act of 2017, represent a belated but significant recognition of mental health as a rights-based public health priority (Srivastava et al., 2016).

Yet a granular, longitudinal understanding of *how* mental health has been debated in India's Parliament—which themes dominate, how discourse has evolved over time, and what policy frames legislators employ—remains elusive. Traditional qualitative methods of legislative analysis, relying on human coding of documents, are inherently limited in scale and susceptible to subjective interpretation. With over two decades of archival Rajya Sabha records now available in digital form, the opportunity to apply computational text analysis to this domain is both timely and academically compelling.

1.2 Problem Statement

Three interrelated challenges motivate the computational approach adopted in this dissertation.

First, parliamentary transcripts constitute an extremely large volume of unstructured text. The Rajya Sabha archive contains thousands of debate records spanning Questions and Answers sessions and plenary debates across twenty-five years. Manual qualitative coding of such a corpus at any meaningful scale is practically infeasible within the constraints of a single research project.

Second, human annotation of thematic content is inherently subjective. Inter-rater reliability is difficult to guarantee across a corpus of this size, and even expert coders may disagree on the boundaries between related policy themes. Computational approaches offer a degree of consistency and reproducibility that manual methods cannot provide.

Third, and most importantly, legislative discourse contains latent topical structures that are not apparent from surface-level reading. A debate nominally about hospital funding may implicitly engage with themes of rural mental health access, workforce shortages, or institutional stigma. Topic models are specifically designed to surface these latent co-occurrence patterns in language (Blei et al., 2003). Computational approaches such as **BERTopic** (Grootendorst, 2022) can therefore be deployed to automatically discover thematic structures within large legislative corpora, providing a scalable, replicable and interpretable framework for parliamentary text analysis.

1.3 Literature Review

Blei, Ng, and Jordan (2003) introduced Latent Dirichlet Allocation (LDA), establishing a foundational generative probabilistic model for modern text analysis. Their framework models documents as mixtures of latent topics, enabling the automated discovery of thematic structures across large, unstructured text corpora without requiring time-consuming manual annotation.

Blei and Lafferty (2006) expanded upon these foundational probabilistic models by introducing Dynamic Topic Models (DTM). Recognizing that thematic discourse is rarely static, this approach incorporates temporal dependencies to track how the internal linguistic structures of topics evolve over time, providing a critical mechanism for longitudinal studies of parliamentary archives.

Reimers and Gurevych (2019) fundamentally shifted text representation by introducing Sentence-BERT (SBERT). Utilizing Siamese and triplet network structures, their architecture generates dense, semantically meaningful sentence embeddings that capture

nuanced contextual relationships between text passages far more effectively than traditional word-frequency models.

Grootendorst (2022) developed BERTopic, a modern framework that integrates state-of-the-art transformer embeddings with clustering algorithms and a class-based TF-IDF procedure. This approach addresses the semantic limitations of earlier probabilistic models by maintaining local semantic relationships and generating highly interpretable topic descriptions specifically suited for complex, nuanced legislative language.

Ristilä and Elo (2023) demonstrated the practical efficacy of computational text analysis by deploying topic modeling against Finnish parliamentary speech datasets. Their longitudinal study successfully identified major political themes and tracked long-term shifts in policy discourse, proving that automated methods can reliably detect underlying societal changes across multiple legislative sessions.

Sharma, Gupta, and Narayan (2021) applied natural language processing specifically to the Indian legislative context by analyzing parliamentary Question Hour sessions. Their research highlights the unique linguistic complexities of Indian political text and demonstrates how automated categorization can map government inquiries, providing a vital stepping stone for analyzing Indian legislative archives.

Srivastava, Chaudhury, and Bhat (2016) provided a foundational qualitative assessment of mental health policy implementation in India. Their substantive policy review argues that despite the severe public health burden, mental healthcare has historically suffered from systemic under-resourcing and a distinct lack of prominent legislative prioritization within broader national healthcare agendas.

Garg, Bhatt, and Patel (2021) intersected computational linguistics with public health by utilizing transformer models to track mental health discourse in formal parliamentary debates. Their automated findings reveal that while the volume of discourse is increasing, mental health is predominantly treated as a secondary, latent theme embedded within broader economic or general health discussions rather than a standalone priority.

Goel (2011) offered critical socio-political context regarding the systemic under-performance of mental health services in low- and middle-income countries like India. By identifying institutional stigma and a pervasive lack of political will as primary barriers, this work helps contextualize why mental health themes often remain latent rather than manifest within decades of legislative text.

1.4 Research Gaps

Methodological Gap: Contextual Representation. Most traditional topic modeling studies rely on Latent Dirichlet Allocation (LDA), which utilizes a bag-of-words repre-

sentation and inherently ignores the contextual relationships between words. Because parliamentary speech consists of complex political discourse, existing probabilistic models like LDA may produce less coherent topics compared to models that preserve semantic nuance.

Technological Gap: Neural Embedding Approaches. There is currently limited research exploring the utility of modern embedding-based topic modeling approaches, such as BERTopic, specifically for the analysis of parliamentary debates. This dissertation addresses this gap by evaluating whether these modern architectures can provide a more scalable and interpretable framework for longitudinal legislative text analysis than traditional methods.

Thematic Gap: Mental Health in Legislative Discourse. Previous computational studies have predominantly focused on general political themes or economic agendas, leaving a significant lack of emphasis on mental health discourse within legislative debates. There is a critical need to determine how mental health is addressed relative to other topics to understand the depth and quality of policy conversations surrounding mental health initiatives.

1.5 Research Objectives

This dissertation pursues the following specific research objectives:

RO1. To design and implement a robust, seven-phase end-to-end data engineering pipeline for harvesting, cleaning, translating, and structuring archival Rajya Sabha debate records pertaining to mental health over the period 2000–2025.

RO2. To apply BERTopic to the curated corpus to identify and interpret latent thematic clusters within India's parliamentary discourse on mental health.

RO3. To conduct a longitudinal analysis of how the salience and framing of mental health-related topics have evolved across parliamentary sessions, with particular attention to periods surrounding major legislative milestones.

Chapter 2

Methodology

2.1 Data Collection and Digitization

2.1.1 Web Scraping and Lexicon Development

The initial phase of corpus construction served as the primary discovery and retrieval mechanism. An automated scraping architecture was developed to systematically query the DSpace-based repository of the Rajya Sabha (*rsdebate.nic.in*). While the repository houses multiple document collections, the extraction protocols were strictly configured to target only formal Plenary Debates, explicitly excluding the shorter Question and Answer sessions to ensure the capture of sustained, substantive policy discourse.

To guarantee a high recall rate of relevant discussions, the retrieval engine utilized an exhaustive, multi-thematic keyword lexicon encompassing clinical terminology, legislative frameworks, historical asylum nomenclature, and socio-cultural impacts. As demonstrated in the simplified algorithmic logic below, the scraper iterated through this lexicon, dynamically building search parameters bounded between the years 2000 and 2025, and parsing the resulting HTML tables to extract critical metadata.

```
Algorithm 1: Plenary Debate Discovery and Metadata Extraction
Input: Lexicon (list of keywords), Target_Collection (Part 2 Debates)
Output: Cleaned_Metadata_Ledger

1: Master_Records <- Empty List
2: For each keyword in Lexicon:
3:     Page_Index <- 0
4:     While True:
5:         URL <- Build_Query(Target_Collection, keyword, Date="2000 TO 2025", Start=
6:         DOM <- Execute_Safe_HTTP_Get(URL)
7:         If DOM contains "no results":
8:             Break inner loop
```

```
9:          Extracted_Rows <- Parse_HTML_Table(DOM)
10:         Master_Records.Append(Extracted_Rows)
11:         Page_Index <- Page_Index + 20
12: Cleaned_Metadata_Ledger <- Remove_Duplicates(Master_Records, subset="Bitstream_Lin
13: Return Cleaned_Metadata_Ledger
```

2.1.2 Dual-Strategy Document Acquisition

A significant operational challenge inherent to archival government databases is the inconsistency of file formats. The Rajya Sabha database contains a heterogeneous mix of digital-first PDFs (machine-readable text) and scanned physical archives. To address this, the pipeline employed a dual-strategy download architecture that parsed the Document Object Model (DOM) of each debate's handle page to locate the appropriate bitstream links.

The acquisition engine searched the web elements using regular expressions to distinguish between native text documents (prefixed with `PD_`) and scanned image counterparts (prefixed with `ID_`). This segregation was critical for computational efficiency downstream. To maintain network stability during the mass acquisition of large files, the downloader incorporated adaptive throttling, comprehensive error-logging, and iterative chunk streaming.

```
Algorithm 2: Resilient Dual-Strategy File Acquisition
Input: Cleaned_Metadata_Ledger
Output: Local Directory of PDF Documents

1: For each Record in Cleaned_Metadata_Ledger:
2:     DOM <- Execute_Safe_HTTP_Get(Record.Bitstream_Link)
3:     Target_PDF_URL <- Null
4:
5:     // Strategy Hierarchy
6:     If DOM contains regex match "/bitstream/.*PD_.*\.pdf":
7:         Target_PDF_URL <- Extract_Link(DOM, "Text PDF")
8:     Else If DOM contains regex match "/bitstream/.*ID_.*\.pdf":
9:         Target_PDF_URL <- Extract_Link(DOM, "Image PDF")
10:
11:    If Target_PDF_URL is not Null:
12:        Stream_Download(Target_PDF_URL, Chunk_Size=8192 bytes)
13:        Log_Success(Record.Session, Record.Date)
14:    Else:
15:        Log_Failure(Record.Bitstream_Link)
16:    Sleep(2 seconds) // Network throttling
```

2.1.3 Optical Character Recognition (OCR) and Text Extraction

For the subset of the corpus consisting of scanned documents, an Optical Character Recognition (OCR) bridge was constructed. Utilizing the `pdf2image` library, each PDF page was first converted into a standardized image array at a resolution of 150 Dots Per Inch (DPI). This specific resolution optimized the balance between character readability for the OCR engine and the severe system memory constraints associated with processing thousands of archival pages.

Text extraction was subsequently executed using the `pytesseract` engine. Because Indian parliamentary discourse frequently code-switches between English and Hindi, standard monolingual OCR configurations yield highly corrupted outputs. Consequently, the OCR engine was explicitly configured with a dual-language parameter to recognize both Latin and Devanagari character sets simultaneously. Given the high probability of memory exhaustion during this intensive computer vision task, a robust checkpointing architecture was implemented, ensuring that partial extractions were saved sequentially to a temporary comma-separated values (CSV) ledger.

```
Algorithm 3: Bilingual OCR with State-Checkpointing
Input: Directory of Scanned PDFs
Output: Raw_Extracted_Text_Ledger

Completed_Files <- Load_JSON_Checkpoint()
For each PDF in Directory:
    If PDF in Completed_Files:
        Continue // Skip previously processed files

    Image_Arrays <- Convert_PDF_to_Images(PDF, DPI=150)
    Document_Text_Blocks <- Empty List

    For each Image in Image_Arrays:
        // Explicitly set language and Page Segmentation Mode (PSM)
        Raw_Text <- Execute_Tesseract(Image, lang="eng+hin", config="--psm 6")
        Paragraphs <- Split(Raw_Text, delimiter="\n")

        For each Para in Paragraphs:
            If Length(Para) > 20 characters:
                Dominant_Lang <- Detect_Language(Para)
                Document_Text_Blocks.Append({Para, Dominant_Lang})

    Append_To_Disk(Document_Text_Blocks, "partial_output.csv")
```

```
20:     Completed_Files.Add(PDF)
21:     Save_JSON_Checkpoint(Completed_Files)
```

2.1.4 Structural Normalization and Multilingual Flagging

Raw data extracted via Optical Character Recognition (OCR) inherently contains severe structural artifacts, including corrupted Unicode characters, non-alphanumeric noise, and truncated words. To accommodate the strict context-window limitations of modern transformer architectures (such as BERT), the raw text was first divided into sequential blocks of a maximum of 500 words, with each chunk inheriting its parent document's metadata.

Following chunking, the text required aggressive heuristic sanitization. However, a critical challenge in parsing Indian parliamentary transcripts is the frequent occurrence of code-switching between English and regional languages. Applying standard non-alphanumeric filters to such text frequently results in the unintended deletion of valid regional scripts. To solve this, a robust "Hindi Protection" and multilingual flagging mechanism was engineered.

The pipeline utilized a two-tiered detection engine to flag the language of every row. The primary tier executed Unicode block analysis, mapping character code points directly to known regional scripts (e.g., `0x0900 - 0x097F` for Devanagari). The secondary tier utilized dictionary-based heuristics to identify Romanized transliterations of Indian languages. Once languages were flagged, structural filters were conditionally applied. Rows flagged as English were subjected to strict thresholding—such as discarding chunks where special characters exceeded 40% or digits exceeded 30%—to eliminate tabular budgetary noise. Conversely, rows flagged as Hindi bypassed these strict filters, preserving the multilingual integrity of the dataset prior to translation. Finally, targeted contextual replacements were executed to repair common domain-specific OCR splits (e.g., mapping "MINIST" to "MINISTER").

```
Algorithm 4: Heuristic Normalization with Regional Script Protection
Input: Raw_Extracted_Text_Ledger
Output: Normalized_Multilingual_Ledger

1: For each Document in Raw_Extracted_Text_Ledger:
2:     Text_Chunks <- Split_Text(Document, max_length=500 words)
3:
4:     For each Chunk in Text_Chunks:
5:         Cleaned_Text <- Remove_Control_Characters(Chunk)
6:         Cleaned_Text <- Apply_Regex_Artifact_Removal(Cleaned_Text)
7:
```

```
        // Two-Tier Language Detection
        If Detect_Unicode_Block(Cleaned_Text) == "Devanagari":
            Lang_Flag <- "hi"
        Else If Contains_Transliteration_Markers(Cleaned_Text):
            Lang_Flag <- "hi"
        Else:
            Lang_Flag <- "en"

        // Conditional Filtering (Hindi Protection)
        If Lang_Flag == "en":
            Special_Char_Ratio <- Calculate_Non_Alnum_Ratio(Cleaned_Text)
            Digit_Ratio <- Calculate_Numeric_Ratio(Cleaned_Text)
            If Special_Char_Ratio > 0.40 OR Digit_Ratio > 0.30:
                Discard Chunk
                Continue

        Normalized_Multilingual_Ledger.Append({Cleaned_Text, Lang_Flag})
```

2.1.5 Machine Translation and Semantic Domain Extraction

Topic modeling algorithms generate the highest semantic cohesion when operating upon a strictly monolingual corpus. Therefore, all non-English text identified in the previous phase required translation. To manage the vast volume of characters and avoid API truncation limits, the text was strategically chunked by sentence boundaries—utilizing periods, exclamation marks, and the Devanagari *Purna Viram* ()—into optimal blocks of 50 to 60 words. This sentence-aware chunking preserved contextual meaning during translation, which arbitrary word-count splits often destroy. The translation was executed using a distributed architecture via the Google Translate API, wrapping calls in a concurrent thread pool equipped with an exponential backoff algorithm to gracefully handle HTTP rate-limiting errors.

Following translation, the corpus contained a massive volume of procedural parliamentary noise, such as role calls, scheduling remarks, and off-topic detours. Passing such noise into an embedding-based topic model like BERTopic severely dilutes the resulting clusters. To isolate the semantic scope, a strict domain-filtering pass was executed using bounded regular expressions.

A comprehensive clinical and policy dictionary was mapped against both the English translation and the original source text to capture all mental health discussions. Crucially, because BERTopic relies on dense sentence embeddings to calculate document distances, input texts must possess sufficient semantic depth; isolated phrases (e.g., “Yes, regarding

mental health") lack the context required for high-dimensional clustering. Therefore, a final mathematical constraint was applied, permanently discarding any extracted row containing fewer than 30 words. This ensured the final dataset consisted exclusively of context-rich, substantive policy debates natively primed for advanced computational linguistics.

```
Algorithm 5: Sentence-Boundary Translation and Domain Extraction
Input: Normalized_Multilingual_Ledger, Mental_Health_Lexicon
Output: Final_BERTopic_Corpus

// Phase A: Machine Translation
For each Row in Normalized_Multilingual_Ledger where Lang != "en":
    Sentence_Blocks <- Regex_Split(Row.Text, delimiters="[.!?]")
    Optimal_Chunks <- Group_Sentences(Sentence_Blocks, target_length=60)

    For each Chunk in Optimal_Chunks:
        Translated_Chunk <- Execute_API_Translation(Chunk) // With exponential bacl
        Row.English_Text.Append(Translated_Chunk)

// Phase B: Domain Extraction and Density Filtering
Regex_Pattern <- Build_Bounded_Pattern(Mental_Health_Lexicon)
Final_BERTopic_Corpus <- Empty List

For each Row in Normalized_Multilingual_Ledger:
    If Regex_Search(Regex_Pattern, Row.English_Text) OR
       Regex_Search(Regex_Pattern, Row.Original_Text):

        Word_Count <- Calculate_Length(Row.English_Text)

        // Semantic density constraint for embedding models
        If Word_Count >= 30:
            Final_BERTopic_Corpus.Append(Row)

Return Final_BERTopic_Corpus
```

2.2 Dynamic Topic Modeling: The BERTopic Architecture

Following the construction and sanitization of the multilingual corpus, the methodological framework transitions from data engineering to computational text analysis. Traditional generative probabilistic models, such as Latent Dirichlet Allocation (LDA), rely heavily on bag-of-words assumptions, which inherently discard the semantic sequencing and deep contextual nuances critical to interpreting legislative discourse. To overcome this limitation, this research deploys BERTopic, a state-of-the-art neural topic modeling framework that executes a sequential pipeline of semantic embedding, dimensionality reduction, density-based clustering, and dynamic temporal tracking.

2.2.1 Semantic Representation via Transformer Embeddings

The foundational step in the BERTopic architecture involves mapping the preprocessed, English-translated parliamentary transcripts into a high-dimensional vector space. This is achieved using a pre-trained Sentence-BERT (SBERT) transformer model. Unlike traditional word-level embeddings, SBERT utilizes Siamese and triplet network structures to derive semantically meaningful sentence embeddings. By processing the text chunks holistically, the model captures the deep, bidirectional context of the policy language, ensuring that semantically similar, yet lexically distinct, phrases (e.g., "psychiatric facility" and "mental asylum") are positioned proximally within the vector space.

2.2.2 Topological Dimensionality Reduction (UMAP)

While transformer embeddings successfully capture deep semantics, their high dimensionality (typically 384 or 768 dimensions) suffers from the "curse of dimensionality," which severely degrades the performance of distance-based clustering algorithms. To mitigate this, the pipeline applies Uniform Manifold Approximation and Projection (UMAP).

UMAP is a nonlinear dimensionality reduction technique founded on Riemannian geometry and algebraic topology. Mathematically, it constructs a high-dimensional fuzzy simplicial complex representation of the data and subsequently optimizes a low-dimensional representation to have the closest possible structural topological equivalence. Unlike Principal Component Analysis (PCA), which strictly preserves global variance, UMAP optimizes the cross-entropy between the high-dimensional and low-dimensional topological representations. This effectively preserves both the local neighborhood relationships (micro-level semantic similarities) and the global data structure (macro-level policy themes), compressing the data into a lower-dimensional space optimized for density extraction.

2.2.3 Density-Based Clustering (HDBSCAN)

Following dimensionality reduction, the framework identifies latent policy themes by clustering the compressed document embeddings. This is executed using Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN).

Traditional algorithms like k-means require the a priori specification of k (the number of topics) and force every document into a cluster, often creating artificial boundaries and degrading topic coherence. HDBSCAN, conversely, operates by defining clusters as highly dense regions separated by sparse regions. It calculates the mutual reachability distance between points, building a minimum spanning tree to extract a stable hierarchy of clusters. Crucially, HDBSCAN is noise-aware; it actively identifies scattered, off-topic parliamentary rhetoric as "outliers" rather than forcing them into a thematic cluster. This mathematically guarantees that the resulting topics are formed exclusively from highly cohesive, substantive policy discussions.

2.2.4 Topic Representation via c-TF-IDF

Once the documents are clustered, the model must extract human-readable topic representations. BERTopic achieves this by treating all documents within a single cluster as a single, concatenated "macro-document" and applying a Class-based Term Frequency-Inverse Document Frequency (c-TF-IDF) procedure. The c-TF-IDF score for a term t within a class (topic) c is calculated as:

$$W_{t,c} = \mathrm{tf}_{t,c} \times \log\left(1 + \frac{A}{f_t}\right) \tag{2.1}$$

Where $\mathrm{tf}_{t,c}$ is the frequency of term t in class c, A is the average number of words per class across the entire corpus, and f_t is the total frequency of term t across all classes. This mathematical formulation penalizes generic parliamentary stop-words that appear universally across all debates while exponentially rewarding terms that are uniquely highly concentrated within a specific thematic cluster, thereby generating highly descriptive and distinct topic labels.

2.2.5 Dynamic Temporal Tracking and Smoothing

To satisfy the longitudinal objectives of this dissertation, the methodology extends beyond static topic extraction to evaluate the temporal evolution of mental health discourse spanning 2000 to 2025. BERTopic facilitates this through dynamic topic modeling, which calculates the c-TF-IDF representations sequentially across designated temporal bins (e.g., legislative years).

Because legislative vocabulary shifts gradually rather than instantaneously, the temporal representations are mathematically smoothed using the representation from the preceding time step. The smoothed weight of term t in class c at time T is calculated as:

$$\vec{v}_{t,c,T} = \alpha\vec{v}_{t,c,T} + (1-\alpha)\vec{v}_{t,c,T-1} \tag{2.2}$$

By tuning the α parameter (typically applying L1 normalization prior to weighting), this temporal smoothing ensures that the resulting longitudinal visualizations reflect genuine, sustained linguistic shifts—such as the gradual transition from archaic terminology like "lunacy" to modern frameworks like "rehabilitation"—rather than artificial, jarring spikes caused by isolated debate sessions.

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